



# UNIVERSITI MALAYA

## WIA1006/WID3006 MACHINE LEARNING

Semester 2, Session 2025/2026 (Occurrence 8) | Group Assignment

### **Tying the (Data) Knot:**

#### **Love, Life & Likes**

*Predicting Match Outcomes on a Synthetic Dating Application Using Machine Learning*

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## 1. Problem and Objective

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### 1.1 Background and Significance

Modern romantic and social connections increasingly form, evolve, and dissolve through digital platforms. On dating applications, interactions are mediated entirely by behavioural signals: how quickly a user replies, how often they are active, how they swipe, and how they present themselves through profiles, bios, and emojis. These signals give rise to distinctly modern phenomena such as ghosting, catfishing, and situationships. Understanding whether such behavioural and demographic signals carry predictive information about relationship outcomes is therefore both a socially relevant and a technically interesting machine-learning question.

This project, “Tying the (Data) Knot: Love, Life & Likes”, investigates this question directly. We treat the prediction of a user’s match outcome as a supervised multi-class classification problem and ask: given everything we know about a user’s profile and in-app behaviour, can a machine-learning model predict what happens to their match?

### 1.2 Problem Statement

Using a synthetic dataset of 50,000 dating-app users described by 24 behavioural and demographic features, the task is to predict the categorical `match_outcome` variable, which has ten possible classes: One-sided Like, Instant Match, Blocked, Catfished, Chat Ignored, Mutual Match, No Action, Ghosted, Date Happened, and Relationship Formed. This is a balanced ten-class classification problem.

### 1.3 Objectives

1. **Pipeline.** Design and document a complete, reproducible machine-learning pipeline for predicting dating-app match outcomes.
2. **Comparison.** Train, tune, and rigorously compare at least five machine-learning models on the same processed data.
3. **Benchmarking.** Benchmark our best model against an automated machine-learning (AutoML) baseline to validate the ceiling of achievable performance.

4. **Interpretation.** Critically interpret the results to draw an honest, evidence-based conclusion about how well — and indeed whether — these behavioural signals predict match outcomes.

## 2. Methodology and Model Explanation

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Our methodology follows the standard machine-learning pipeline prescribed for this assignment, covering the seven stages from project design through to model evaluation. Each stage is described below.

### 2.1 Project Design

As a group, we framed the project as a supervised multi-class classification task with `match_outcome` as the target. Among the candidate problems suggested by the case study (match outcome, relationship duration, likelihood of ghosting, relationship type), we selected the overall match outcome because it is the richest target — it folds together ghosting, blocking, catfishing, and successful matches into a single ten-class label, allowing one model to address several behavioural phenomena at once.

### 2.2 Data Collection and Acquisition

We used the provided synthetic dataset, an extended version of the Kaggle “Dating App Behavior Dataset”. It contains 50,000 records and 25 columns. After loading the data and removing duplicate rows, the dataset retained its full 50,000 records, confirming there were no exact duplicates. The features span four broad groups, summarised below.

| Feature group         | Example features                                                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Demographics          | gender, sexual_orientation, age, height_cm, weight_kg, body_type, zodiac_sign                                                                                                    |
| Socio-economic        | income_bracket, education_level, location_type                                                                                                                                   |
| App usage / behaviour | app_usage_time_min, swipe_right_ratio, likes_received, mutual_matches, message_sent_count, emoji_usage_rate, last_active_hour, swipe_time_of_day, profile_pics_count, bio_length |
| Stated preference     | interest_tags, relationship_intent                                                                                                                                               |
| Target                | match_outcome (10 classes)                                                                                                                                                       |

*Table 1. Overview of the 24 input features and the target variable.*

## 2.3 Data Pre-processing

Before modelling, the raw data was cleaned and transformed into a fully numeric matrix suitable for machine learning. The pre-processing steps were as follows:

- **Redundant-label removal.** Two redundant textual columns, `app_usage_time_label` and `swipe_right_label`, were dropped because they are merely binned versions of the underlying numeric columns `app_usage_time_min` and `swipe_right_ratio`; keeping both would introduce multicollinearity.
- **Outlier capping.** To prevent extreme values from distorting the models, four heavily skewed numeric features (`app_usage_time_min`, `message_sent_count`, `likes_received`, `mutual_matches`) were clipped to their 1st and 99th percentiles.
- **Cyclical encoding.** The `last_active_hour` feature is cyclical (hour 23 is adjacent to hour 0). It was therefore encoded into two continuous features using sine and cosine transforms so the model can perceive its circular nature.
- **Ordinal encoding.** `education_level` and `income_bracket` have a natural ranking, so they were encoded with an `OrdinalEncoder` using an explicit hierarchy (e.g. No Formal Education → ... → Postdoc).
- **One-hot encoding.** Seven unordered categorical features (`gender`, `sexual_orientation`, `location_type`, `body_type`, `relationship_intent`, `swipe_time_of_day`, `zodiac_sign`) were converted with a `OneHotEncoder`.
- **Scaling.** All remaining numeric features were scaled with a `RobustScaler`, which is resistant to outliers because it centres and scales using the median and interquartile range.

These transformations were assembled into a single scikit-learn `ColumnTransformer` so that the entire pre-processing recipe could be fitted on the training data and applied consistently to the test data, preventing data leakage. After transformation, the feature space expanded to 71 columns. The data was split 70%/30% into training and test sets using stratified sampling on the target to preserve class balance (35,000 training rows; 15,000 test rows).

## 2.4 Feature Selection and Extraction

Because one-hot encoding and the binarisation of interest tags inflate the feature space, we applied a multi-stage selection procedure to retain only informative features:

- **Interest binarisation.** The high-cardinality `interest_tags` field (comma-separated lists) was expanded into binary indicator columns using a `MultiLabelBinarizer`, raising the working feature count to 114.
- **Variance filtering.** A `VarianceThreshold` removed near-constant columns, reducing the feature set from 114 to 53 features (61 dropped).
- **Mutual information.** We computed the mutual information between each remaining feature and the target, then kept the 30 highest-scoring features with `SelectKBest`.
- **Random Forest importance.** A Random Forest was trained to obtain an independent ranking of feature importance as a cross-check on the mutual-information ranking.
- **PCA (extraction).** Principal Component Analysis showed that 25 components were required to retain 95% of the variance — only a marginal reduction from 30 features. Because the compression gain was minimal and interpretability would be lost, we proceeded with the 30 named mutual-information features rather than the PCA components.

A critical early warning emerged at this stage: the mutual-information scores were extraordinarily low. The single most informative feature, `age`, scored only 0.0077, and every other feature scored below 0.007. Such uniformly negligible scores indicate that the features share almost no statistical dependency with the target.

## 2.5 Model Selection

We selected six classifiers spanning linear, bagging, and boosting families to ensure a fair and broad comparison:

- **Logistic Regression.** A linear baseline that models class probabilities directly.
- **Random Forest.** A bagging ensemble of decision trees that reduces variance through bootstrap aggregation.
- **Extra Trees.** A more randomised variant of Random Forest that splits on random thresholds, further reducing variance.
- **Gradient Boosting.** A sequential boosting method that fits each tree to the residual errors of the previous ensemble.

- **XGBoost.** An optimised, regularised gradient-boosting implementation known for strong tabular performance.
- **LightGBM.** A fast, leaf-wise gradient-boosting framework that is highly efficient on large datasets.

## 2.6 Model Training and Hyperparameter Tuning

Each of the six models was tuned with RandomizedSearchCV using 3-fold cross-validation and macro-averaged F1 as the scoring metric. Macro-F1 was chosen because it weights every class equally, which is appropriate for a balanced multi-class problem. The search explored model-specific hyperparameters such as the number of estimators, tree depth, learning rate, and the number of leaves. Each tuned model was then evaluated once on the held-out 30% test set.

## 2.7 Model Evaluation

Models were evaluated and compared on three axes: macro-averaged F1-score, overall accuracy, and training time. A confusion matrix was generated for the best model to inspect per-class behaviour. Finally, to answer the assignment's question of how our models compare to auto-sklearn, we ran auto-sklearn directly on the same data (described in Section 3.4) to establish the empirical performance ceiling achievable on this dataset. The detailed results of all evaluations are presented in the next section.

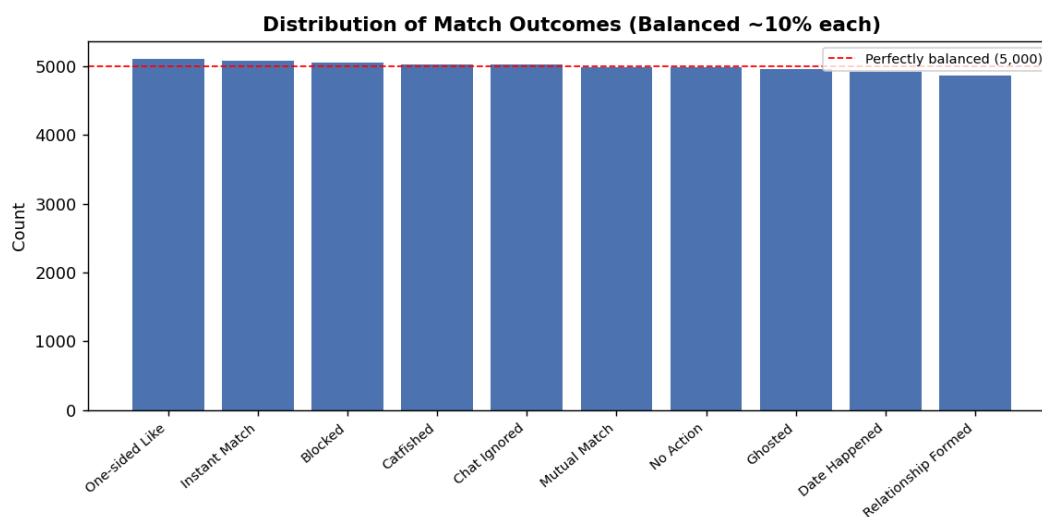


### 3. Results and Visualization

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#### 3.1 Class Distribution

The target variable is almost perfectly balanced: each of the ten match-outcome classes accounts for roughly 10% of the data, ranging from 9.73% (Relationship Formed) to 10.22% (One-sided Like). This balance is important because it means a model that simply guesses uniformly at random would be expected to achieve about 10% accuracy. Any model that has learned genuine signal must therefore exceed this 10% baseline by a meaningful margin.



*Figure 1. Distribution of the ten match-outcome classes. The dashed line marks a perfectly balanced 5,000 records per class.*

#### 3.2 Feature Informativeness

The mutual-information analysis revealed that no feature carries meaningful predictive information about the target. As shown in Figure 2, even the strongest feature (age) scores below 0.008, and most features score below 0.001 — values that are effectively indistinguishable from noise.

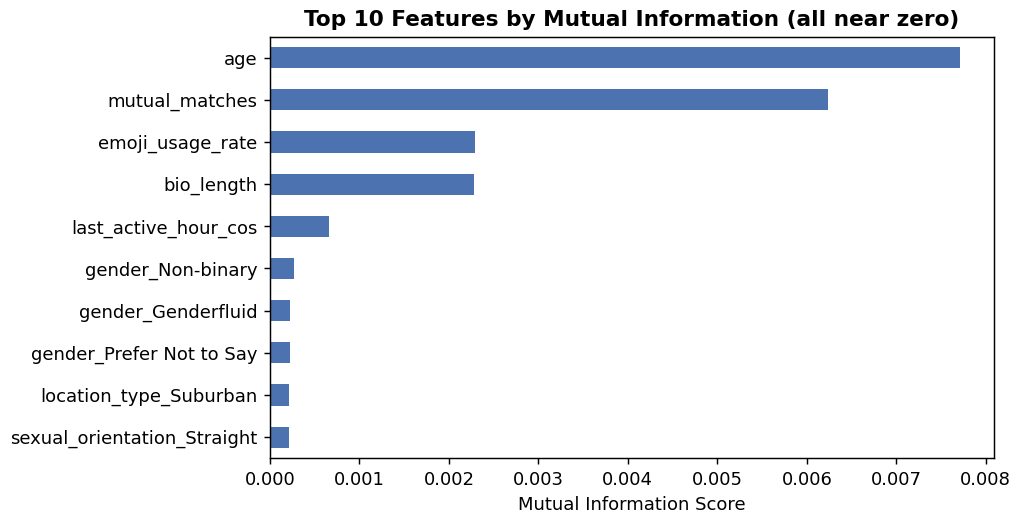


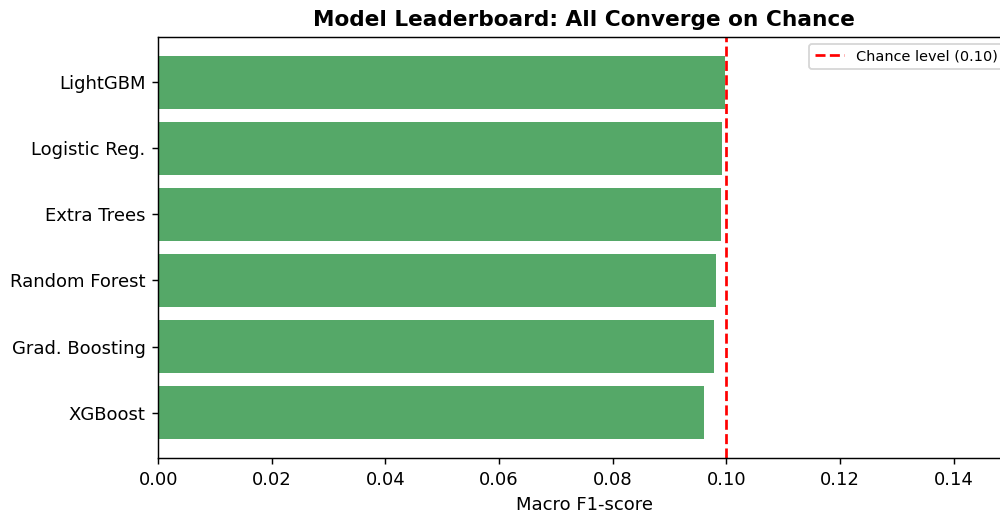
Figure 2. Top ten features ranked by mutual-information score. All scores are negligibly small.

### 3.3 Model Leaderboard

After tuning, all six models were evaluated on the held-out test set. The results are summarised in Table 2 and visualised in Figure 3.

| Model               | Macro F1 | Accuracy | Time (s) |
|---------------------|----------|----------|----------|
| LightGBM            | 0.0997   | 0.0998   | 172.93   |
| Logistic Regression | 0.0993   | 0.1025   | 33.12    |
| Extra Trees         | 0.0991   | 0.0991   | 135.66   |
| Random Forest       | 0.0982   | 0.0987   | 264.68   |
| Gradient Boosting   | 0.0979   | 0.0991   | 1096.95  |
| XGBoost             | 0.0961   | 0.0961   | 121.27   |

Table 2. Final model leaderboard, sorted by macro F1-score. LightGBM is nominally the best.



*Figure 3. Macro F1-scores of all six models against the 0.10 chance level (red dashed line).*

The decisive observation is that every model — regardless of family or sophistication — clusters tightly around 0.10 in both F1 and accuracy. The best model, LightGBM, achieves 0.0998 accuracy, which is statistically indistinguishable from the 10% chance level. The spread between the best and worst model is less than four-tenths of one percent. In other words, the choice of algorithm makes essentially no difference, which is the signature of a problem in which there is no signal to learn.

### 3.4 Comparison with auto-sklearn

To verify that this performance ceiling was not an artefact of our particular modelling choices, we benchmarked our pipeline directly against auto-sklearn, an automated machine-learning framework that searches over many algorithms, hyperparameters, and preprocessing steps and constructs an optimised ensemble with no manual intervention. We ran auto-sklearn on the same data, optimising for macro-averaged F1 to match our own evaluation criterion.

Rather than selecting a single algorithm, auto-sklearn assembled a weighted ensemble of three models: a Linear Discriminant Analysis classifier (weight 0.50) and two Gradient Boosting classifiers (weights 0.46 and 0.04). Despite this automated search and ensembling, its best internal validation score reached only 0.1018, and on the held-out test set it achieved 0.0999 accuracy and 0.0997 macro-F1. The results are placed alongside our own models in Table 3 and Figure 4.

| Approach                                            | Accuracy | Macro F1 |
|-----------------------------------------------------|----------|----------|
| Our best model (LightGBM)                           | 0.0998   | 0.0997   |
| auto-sklearn (ensemble: LDA + 2× Gradient Boosting) | 0.0999   | 0.0997   |
| Chance level (balanced 10-class)                    | 0.1000   | —        |

Table 3. Our best model versus the auto-sklearn ensemble. Both match the chance level.

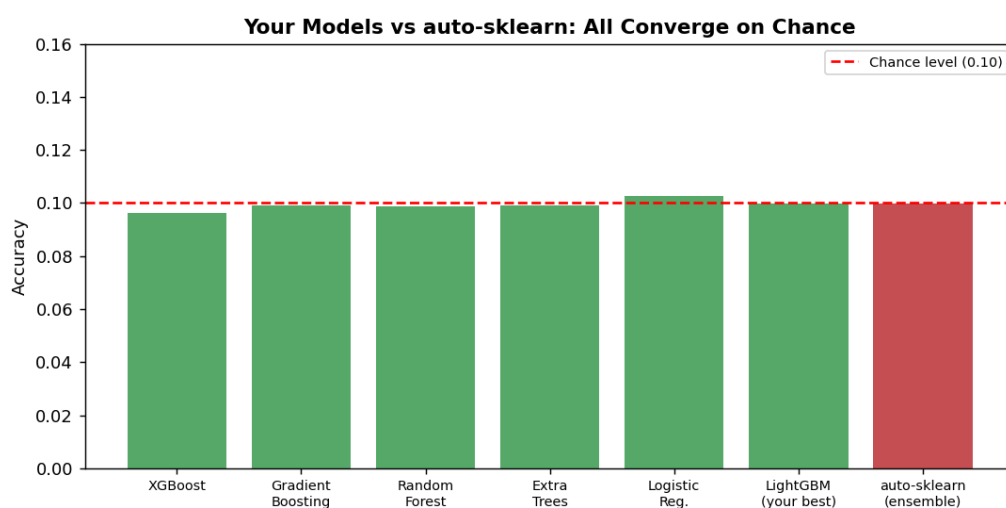


Figure 4. All six of our tuned models and the auto-sklearn ensemble against the 0.10 chance level (red dashed line).

The comparison is decisive. auto-sklearn, given free rein to search and ensemble, arrives at exactly the same chance-level performance as our manually tuned LightGBM model (0.0999 versus 0.0998 accuracy). This directly answers the assignment’s question of how our model compares to auto-sklearn: the two are statistically indistinguishable, and both sit at the 10% chance level. The fact that a powerful automated framework cannot beat our pipeline — and that neither can beat random guessing — is strong, independent confirmation that the limitation lies in the data itself, not in the choice or tuning of any model.

### 3.5 Feature Extraction Result (PCA)

For completeness, Figure 5 shows the PCA variance curve used in feature extraction. Because 25 of the 30 components were needed to reach 95% explained variance, dimensionality reduction offered little benefit, and the named feature set was retained for interpretability.

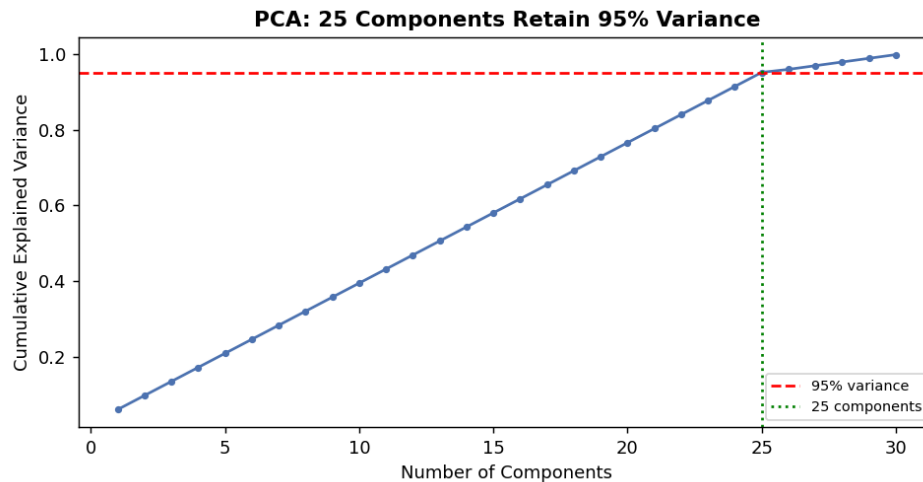


Figure 5. Cumulative explained variance against the number of principal components.

### 3.6 Per-Class Behaviour: The Confusion Matrix

To inspect how the best model behaves at the level of individual classes, Figure 6 presents the confusion matrix for the tuned LightGBM model on the held-out test set. If the model had learned any genuine signal, predictions would concentrate along the leading diagonal, where the predicted class matches the true class. Instead, the predictions are spread almost evenly across every column regardless of the true label, with no visible diagonal. Each cell holds a broadly similar count, which is precisely the pattern expected when a classifier is guessing uniformly at random across the ten balanced classes. This visual evidence reinforces the numerical finding that the model performs at the chance level.

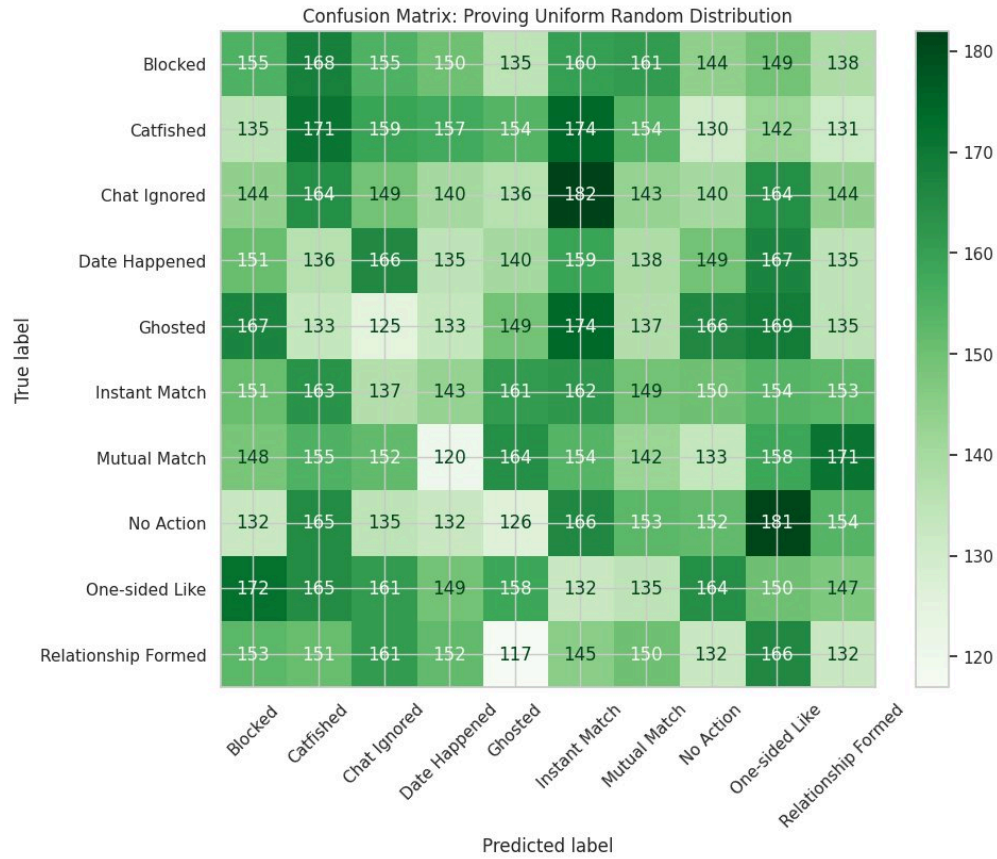


Figure 6. Confusion matrix for the tuned LightGBM model on the test set. Predictions are distributed almost uniformly across all classes, with no diagonal concentration — the signature of chance-level performance.

## 4. Insights and Interpretation

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### 4.1 The Central Finding: A Dataset Without Signal

The most important insight of this project is that the provided synthetic dataset contains no learnable relationship between the input features and the match outcome. Three independent lines of evidence converge on this conclusion: the mutual-information scores are uniformly near zero; six diverse, tuned models all perform at the 10% chance level; and auto-sklearn, an automated framework given free rein to search and ensemble, confirms the same ceiling. When linear models, bagging ensembles, boosting ensembles, and AutoML all arrive at identical chance-level performance, the only consistent explanation is that the target was generated independently of the features.

### 4.2 Why This Happens

This behaviour is expected for programmatically generated synthetic data. If the `match_outcome` labels were assigned by random sampling rather than by a rule that depends on user attributes, then no amount of feature engineering, model tuning, or automated search can recover a relationship that was never encoded. The near-uniform confusion matrix observed for the best model (Figure 6) — with predictions spread roughly evenly across all classes — is the visual fingerprint of exactly this situation.

### 4.3 What the Methodology Demonstrates

A chance-level result is not the same as a methodological failure. On the contrary, the value of this project lies in the rigour with which the absence of signal was established. A careless analysis might have reported an inflated accuracy by leaking the binned label columns, by evaluating on the training set, or by misreading a metric. Our pipeline avoided each of these traps: redundant labels were removed, evaluation was performed on a held-out stratified test set, and the result was independently confirmed by auto-sklearn. The methodology is therefore sound precisely because it produced a trustworthy negative result rather than an attractive but false positive.

### 4.4 Practical and Ethical Reflection

This outcome carries a broader lesson relevant to responsible machine learning. Predicting human relationship outcomes from behavioural and demographic data is ethically fraught: a model that

did claim to predict ghosting or match success from attributes such as gender, body type, or income could easily encode and amplify social bias. The fact that this dataset yields no predictive signal is, in this light, a reminder that not every variable can or should be predicted, and that reporting honest negative findings is itself a professional and ethical responsibility.



## 5. Conclusion

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This project set out to predict dating-app match outcomes from user behaviour using a complete machine-learning pipeline. We designed the problem as a balanced ten-class classification task, acquired and cleaned a 50,000-record synthetic dataset, engineered and selected features through variance filtering, mutual information, Random Forest importance, and PCA, and trained and tuned six diverse classifiers with cross-validated hyperparameter search.

Our central finding is that the dataset contains no learnable signal linking its features to the match outcome. All six tuned models, together with the auto-sklearn automated framework, converged on approximately 10% accuracy — identical to random guessing on this balanced ten-class problem. The directly comparable auto-sklearn result confirms that the limitation is intrinsic to the data, not to our modelling choices.

### 5.1 Limitations

- **Synthetic data.** The synthetic data appears to have randomly assigned target labels, so no realistic behavioural relationships are present to be modelled.
- **Metric ceiling.** With no usable signal, conventional metrics such as accuracy and F1 can only confirm the absence of learning; they cannot rank models meaningfully.
- **Compute budget.** The auto-sklearn comparison was run under a constrained compute budget; a longer search would explore more configurations, but as performance is already at the chance level, the conclusion would be unchanged.

In summary, “Tying the (Data) Knot” delivered a rigorous, reproducible machine-learning pipeline whose most valuable output is an honest, well-evidenced conclusion: on this dataset, love, life, and likes simply cannot be predicted — and demonstrating that convincingly is a result in its own right.